

Time Investment Tracker

Vision

A personal productivity application that helps users understand how they invest their time across learning, work, and personal activities.

The application focuses on answering:

- Where did my time go this week?
- Am I spending time according to my priorities?
- Which activities are growing or declining over time?
- How does this week compare to previous weeks?

The application is local-first and supports synchronization between desktop and mobile devices without relying on external cloud providers.

Product Principles

Local First

All data must be stored locally.

The application must remain fully functional without internet access.

User Ownership

The user owns all data.

Data should be exportable at any time.

Simple Tracking

Recording time must require minimal effort.

A user should be able to start or stop a session in one click.

Insights Over Raw Data

Reports should help users make decisions rather than simply display numbers.

Core User Journey

Create Activity

Examples:

- English
 - System Design
 - Reading
 - Podcast
 - Coding
 - Exercise
-

Start Session

User selects an activity.

Start timer.

Stop Session

Timer stops.

Session is saved.

Example:

Activity: English

Duration: 60 minutes

Date: 2026-06-10

Device: Mobile

Review Reports

User can review:

- Daily report

- Weekly report
 - Monthly report
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Domain Model

Activity

Represents a category of time investment.

Properties:

- id
- name
- color
- icon
- weeklyGoalMinutes
- createdAt

Examples:

- English
 - Reading
 - Coding
-

Session

Represents a tracked period of time.

Properties:

- id
- activityId
- startTime
- endTime
- durationMinutes
- deviceId
- notes
- createdAt

Rules:

- Duration must be greater than zero.
- Session belongs to exactly one activity.

- Session cannot overlap itself.
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Weekly Goal

Represents a target amount of time.

Properties:

- id
- activityId
- targetMinutes

Example:

English

Target = 600 minutes

10 hours per week

Report

Generated from session data.

Not stored permanently.

Calculated on demand.

Types:

- Daily
 - Weekly
 - Monthly
 - Yearly
-

Business Rules

Activity Rules

- Activity name must be unique.
- Activity cannot be deleted if sessions exist.

- Activity can be archived.
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Session Rules

- Session duration must be positive.
 - Running session must be unique.
 - Only one active timer can exist at a time.
 - Stopping a timer automatically calculates duration.
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Goal Rules

- Goal must be greater than zero.
 - Goal measured in minutes.
 - Goal can be changed at any time.
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Reporting Rules

Weekly report starts on Monday.

Monthly report uses calendar month.

All reports are generated from sessions.

No aggregated totals are stored.

Synchronization Strategy

Architecture

Mobile App

Desktop App

Local Database

Synchronization Layer

Sync Unit

The smallest syncable item is Session.

Never sync totals.

Always sync raw events.

Good:

Session A = 60 min

Session B = 45 min

Bad:

English Total = 105 min

Conflict Resolution

Each record contains:

- id
- createdAt
- updatedAt

Last write wins.

Deleted records are soft deleted.

Technical Architecture

Frontend

Flutter

Targets:

- Android
- Windows

Future:

- Linux
 - Web
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State Management

Riverpod

Local Database

SQLite

ORM:

Drift

Project Layers

presentation

application

domain

infrastructure

Folder Structure

lib/

core/

shared/

features/

activity/

session/

report/

settings/

domain/

application/

presentation/

infrastructure/

Feature Roadmap

Phase 1

Activity Management

Session Tracking

Start Timer

Stop Timer

Weekly Report

Monthly Report

SQLite Storage

Phase 2

Goals

Progress Tracking

Charts

Weekly Review

Monthly Review

Phase 3

Synchronization

Desktop Sync

Mobile Sync

Conflict Resolution

Backup

Restore

Phase 4

Advanced Insights

Trends

Focus Analysis

Learning Analytics

Priority Suggestions

Success Metrics

The application is successful if:

- User tracks time daily.
- User reviews weekly reports.
- User changes priorities based on reports.
- User can clearly identify where time is spent.

Primary metric:

Percentage of planned time versus actual time.

Example:

English Goal = 10 hours

Actual = 8 hours

Completion = 80%